

API TESTING REPORT

Input Validation with FluentValidation — Day 4

Postman-Based Testing of the TaskTrackerApi Validation Layer

1. Introduction

This report documents the functional testing of the input validation layer added to the TaskTrackerApi on Day 4, building on the authorization and role-based access control implemented on Day 3. FluentValidation was registered and connected to the API pipeline, and testing focused on verifying that invalid requests to the Create and Update Task endpoints are rejected before reaching controller logic, with structured, field-level error responses.

All requests were executed manually in Postman against the Create Task and Update Task endpoints, each tested with a range of invalid payloads designed to trigger a specific validation rule.

2. Objective

- Confirm that TaskCreateDtoValidator rejects invalid Create Task payloads with 400 Bad Request.
- Confirm that TaskUpdateDtoValidator rejects invalid Update Task payloads with 400 Bad Request.
- Verify that validation runs automatically during model binding, before controller logic executes.
- Verify that error responses follow the structured ValidationProblemDetails format.
- Document evidence of each test case, including the field-level validation message returned.

3. Testing Environment

Project	TaskTrackerApi
Technology	FluentValidation, ASP.NET Core Model Binding, ValidationProblemDetails
Testing Tool	Postman (manual requests, one request per validation rule)
Validators Under Test	TaskCreateDtoValidator, TaskUpdateDtoValidator
Date	August 11, 2026

4. Test Cases — Input Validation

The table below lists all test cases executed against the Create Task and Update Task endpoints. Each case sends a payload violating exactly one validation rule and confirms the API returns 400 Bad Request with a structured error naming the offending field.

TC	Test Case	Endpoint	Rule Violated	Expected	Result
TC-01	Create Task — Empty Title	POST /api/Tasks	Title is required	400	PASS
TC-02	Create Task — Title Contains Only Spaces	POST /api/Tasks	Title must not be whitespace	400	PASS
TC-03	Create Task — Title Exceeds Maximum Length	POST /api/Tasks	Title must not exceed 200 characters	400	PASS
TC-04	Create Task — Invalid UserId	POST /api/Tasks	UserId must be valid / must exist	400	PASS
TC-05	Update Task — Empty Title	PUT /api/Tasks/1	Title is required	400	PASS
TC-06	Update Task — Title Contains Only Spaces	PUT /api/Tasks/1	Title must not be whitespace	400	PASS
TC-07	Update Task — Title Exceeds Maximum Length	PUT /api/Tasks/1	Title must not exceed 200 characters	400	PASS

5. Detailed Test Case Evidence

Each test case below includes the endpoint, the rule under test, the expected outcome, and a dedicated space to attach the Postman screenshot (request payload and 400 response) captured during execution.

TC-01 Create Task — Empty Title

Endpoint	POST /api/Tasks
Rule Violated	Title is required
Payload Condition	Title field submitted as an empty string.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	"Title" must not be empty.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Task Tracker API Version2 / Validation Rule / TC-01 | Create Task - Empty Title

POST

{{baseUrl}}/api/tasks

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

```
1 {
2   "title": "",
3   "userId": "6ee135d1-4ac7-4783-94c4-a65766715654"
4 }
```

Body

Cookies

Headers (4)

Test Results (1/1)

400 Bad Request

53 ms

315 B

Save Response

JSON

Preview

Debug with AI

```
1 {
2   "title": "One or more validation errors occurred.",
3   "status": 400,
4   "errors": {
5     "Title": [
6       "'Title' must not be empty.",
7       "Title cannot contain only spaces."
8     ]
9   }
10 }
```

TC-02 Create Task — Title Contains Only Spaces

Endpoint	POST /api/Tasks
----------	-----------------

Rule Violated	Title must not be whitespace
Payload Condition	Title field submitted containing only space characters.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	Title cannot be empty or whitespace.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

[HTTP](#) Task Tracker API Version2 / Validation Rule / TC-02 Create Task - Title Contains Only Spaces

Save

Share

POST

{{baseUrl}}/api/tasks

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Auth Type

Be...

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results (1/1)

400 Bad Request

6 ms

315 B

Save Response

{}

JSON

Preview

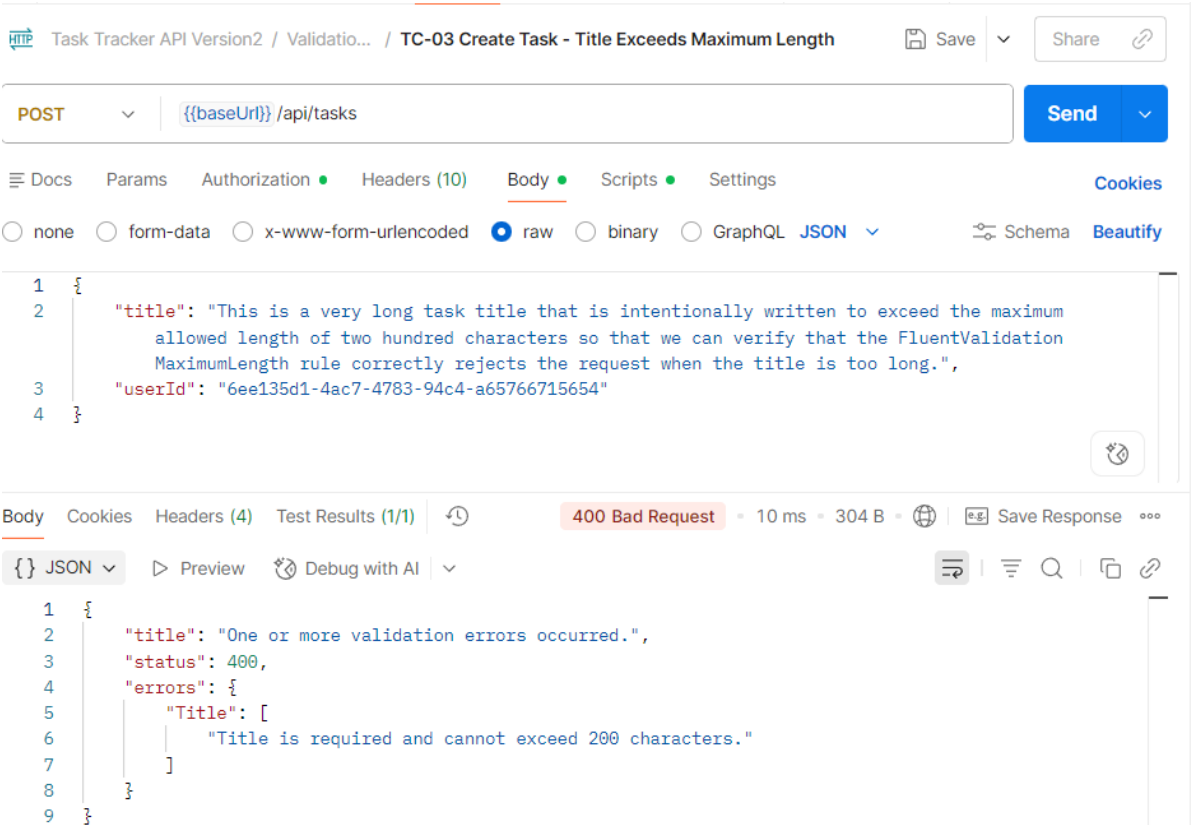
Debug with AI

```
1  {
2    "title": "One or more validation errors occurred.",
3    "status": 400,
4    "errors": {
5      "Title": [
6        "'Title' must not be empty.",
7        "Title cannot contain only spaces."
8      ]
9    }
10 }
```

TC-03 Create Task — Title Exceeds Maximum Length

Endpoint	POST /api/Tasks
Rule Violated	Title must not exceed 200 characters
Payload Condition	Title field submitted with a string longer than 200 characters.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	Title cannot exceed 200 characters.
Actual Result	400 Bad Request — PASS

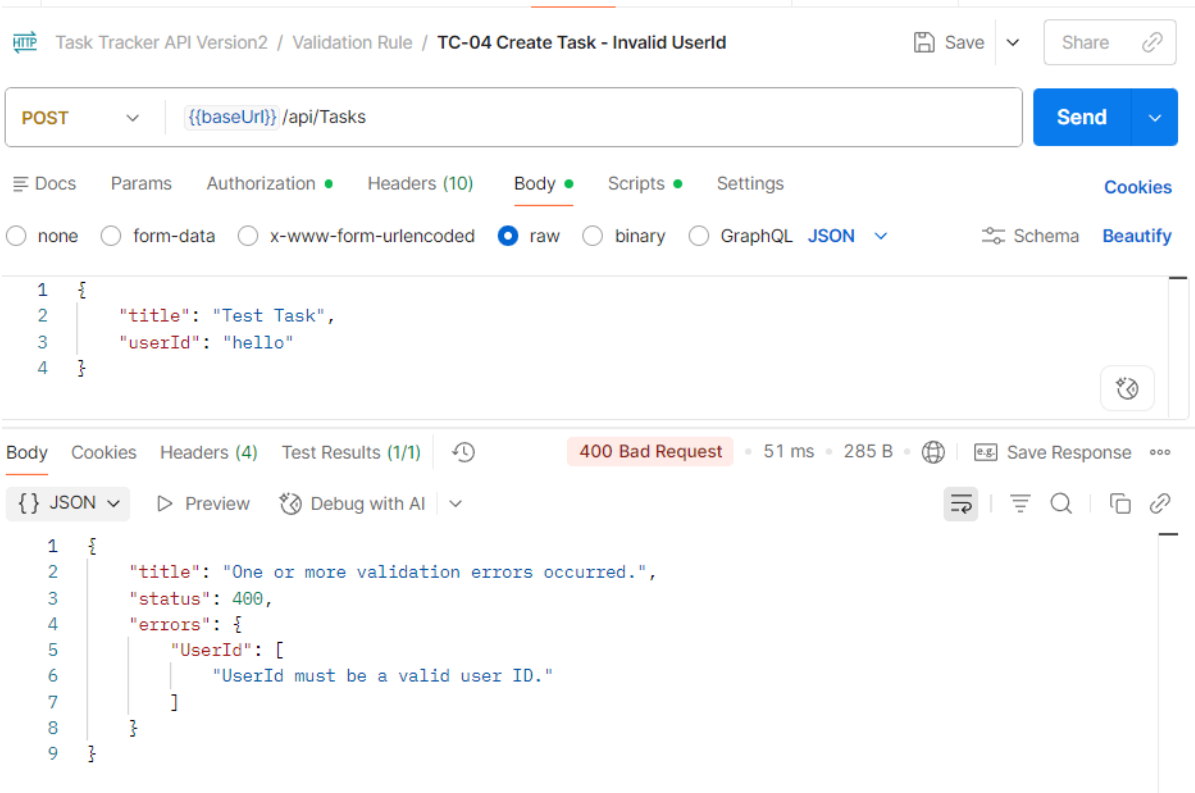
Evidence / Screenshot



TC-04 Create Task — Invalid UserId

Endpoint	POST /api/Tasks
Rule Violated	UserId must be valid / must exist
Payload Condition	UserId field submitted referencing a non-existent user.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	UserId must reference an existing user.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot



TC-05 Update Task — Empty Title

Endpoint	PUT /api/Tasks/{id}
Rule Violated	Title is required
Payload Condition	Title field submitted as an empty string.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	"Title" must not be empty.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

[HTTP](#) Task Tracker API Version2 / Validation Rule / TC-05 Update Task - Empty Title

PUT

{{baseUrl}}/api/tasks/1

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

```
1 {
2   "title": "   ",
3   "isCompleted": false
4 }
```

Body

Cookies

Headers (4)

Test Results (1/1)

400 Bad Request

11 ms

307 B

Save Response

JSON

Preview

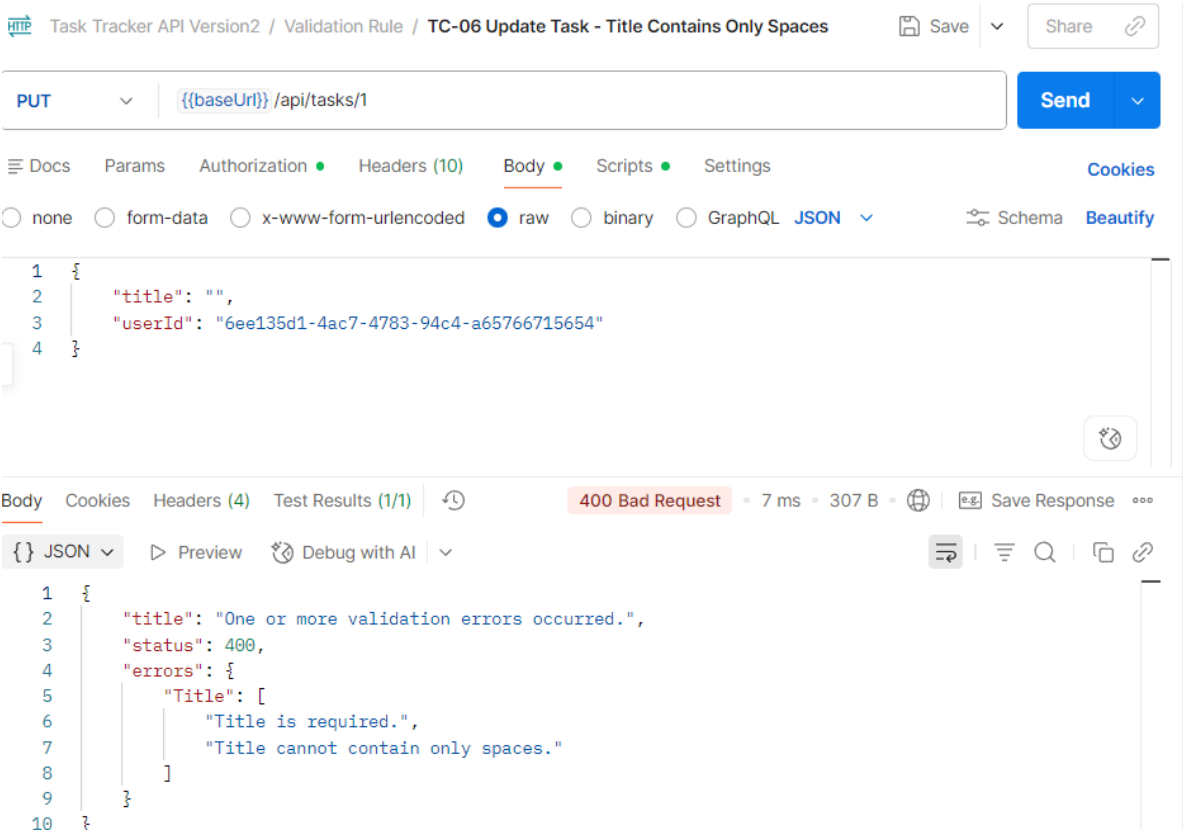
Debug with AI

```
1 {
2   "title": "One or more validation errors occurred.",
3   "status": 400,
4   "errors": {
5     "Title": [
6       "Title is required.",
7       "Title cannot contain only spaces."
8     ]
9   }
10 }
```

TC-06 Update Task — Title Contains Only Spaces

Endpoint	PUT /api/Tasks/{id}
Rule Violated	Title must not be whitespace
Payload Condition	Title field submitted containing only space characters.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	Title cannot be empty or whitespace.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot



TC-07 Update Task — Title Exceeds Maximum Length

Endpoint	PUT /api/Tasks/{id}
Rule Violated	Title must not exceed 200 characters
Payload Condition	Title field submitted with a string longer than 200 characters.
Expected Result	400 Bad Request — ValidationProblemDetails
Sample Error Message	Title cannot exceed 200 characters.
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Task Tracker API Version2 / Validation... / TC-07 Update Task - Title Exceeds Maximum Length Save Share

PUT ▼ `{{baseUrl}}/api/tasks/1` Send ▼

Docs Params Authorization Headers (10) Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼ Schema Beautify

```
1 {
2   "title": "This is a very long task title that is intentionally written to exceed the maximum
3     allowed length of two hundred characters so that we can verify that the FluentValidation
4     MaximumLength rule correctly rejects the request when the title is too long.",
5   "isCompleted": false
6 }
```

Body Cookies Headers (4) Test Results (1/1) 400 Bad Request = 4 ms = 288 B = Save Response ...

{} JSON ▼ Preview Debug with AI ▼ ↩ ≡ 🔍 📄 🔗

```
1 {
2   "title": "One or more validation errors occurred.",
3   "status": 400,
4   "errors": {
5     "Title": [
6       "Title cannot exceed 200 characters."
7     ]
8   }
9 }
```

6. Sample Structured Error Response

TC-03 and TC-07 (title exceeds maximum length) both returned a `ValidationProblemDetails` body naming the field and the specific rule that failed:

```
{
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "Title": [
      "Title cannot exceed 200 characters."
    ]
  }
}
```

7. Test Summary

Total Test Cases	7
Create Task Validation Tests	4
Update Task Validation Tests	3
Passed	7
Failed	0

8. Conclusion

All 7 test cases executed against the validation layer returned the expected 400 Bad Request status, confirming that `FluentValidation` intercepts invalid payloads during model binding before any controller action executes. Both the Create and Update validators correctly rejected empty titles, whitespace-only titles, over-length titles, and — for Create — an invalid `UserId`, with no requests reaching business logic on an invalid payload.

Every failure response followed the structured `ValidationProblemDetails` shape, naming the specific field and rule that failed rather than returning a generic error string. This validates the validation foundation built today and confirms the API is ready to have further business-rule validators (e.g. cross-field checks such as end-date-after-start-date) added in future iterations.

— end of Day 4 Postman Validation Testing Report —